MAD: Multimodal Framework for Adaptive Time-Series Anomaly Detection

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***Abstract*—Pre-trained time-series anomaly detector TSPulse relies on multiple specialized heads, yet picks a single static head per dataset, which is less effective for diverse series and infeasible on unseen datasets. We propose MAD, a two-stage framework that (i) performs selective dimensionality reduction, using best-channel selection when a consistent channel scheme and a small tuning set exist or PCA otherwise, and (ii) uses a Large Multimodal Model (LMM) to visually select the most appropriate head per channel with a few-shot, dataset-agnostic prompt. On the TSB-AD-M benchmark, MAD attains a mean VUS-PR of 0.437, improving over TSPulse (ZS 0.361) and xLSTMAD (0.37) without gradient-based training, while remaining competitive with a static *triangulation* rule (0.442) that presupposes labeled tuning data and dataset identity. Ablations show the reduction stage is essential for practicality, cutting runtime by >93% and API cost by >94% relative to scoring every channel. These results indicate that visual, dataset-agnostic head selection is a robust alternative to static rules, especially when prior dataset knowledge is unavailable. The source code is available at https://github.com/seokjeongeum/MAD.**

***Index Terms*—Time-series anomaly detection, large multimodal models (LMMs), dynamic model selection, dimensionality reduction, pre-trained models**

I. Introduction

Time-series anomaly detection is the task of identifying data points or patterns in sequential data that deviate significantly from what is considered normal [1]. This is a vital task in nearly every scientific and industrial field, with applications ranging from spotting fraudulent credit card transactions to monitoring a patient's vital signs in healthcare and detecting faults in industrial machinery [2]–[4]. Much of this data is multivariate. Formally, a multivariate time series is a sequence of observations $X = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_T\}$ of length T, where each observation $\mathbf{x}_t \in \mathbb{R}^N$ is a vector of N values recorded at time t. Each of the N variables is referred to as a channel [5]. Given this data, the goal of time-series anomaly detection is to take X as input and produce a corresponding anomaly score sequence $\mathbf{a} = \{a_1, a_2, \ldots, a_T\}$, where a higher score a_t indicates a higher likelihood that the observation $\mathbf{x}_t$ is anomalous [6].

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However, time-series data often contains intricate seasonalities that constitute normal behavior [7]. Distinguishing a genuine anomaly from these complex normal variations is a major hurdle. To tackle this complexity, the field has moved from classical statistical methods toward more powerful learning-based approaches, such as the TSPulse model [8]. These models, which have demonstrated state-of-the-art performance on modern benchmarks, can learn intricate patterns directly from data, enabling them to build more accurate representations of normal behavior.

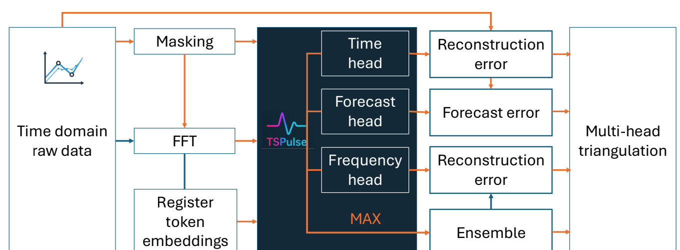

Fig. 1. The anomaly detection workflow in TSPulse. The framework employs self-supervised learning by masking raw data and training specialized 'heads' to reconstruct it from both time and frequency perspectives. The final anomaly score is derived from the reconstruction or forecast errors.

TSPulse is built on the insight that different types of anomalies are best spotted from different perspectives. To capture this, it uses a *multi-head* architecture: a suite of specialized detectors (heads) that analyze the data from complementary viewpoints, as illustrated in Fig. 1. For example, a **Time Head** reconstructs the data in the time domain, while a **Frequency Head** works in the frequency domain, and a **Forecast Head** predicts future values to identify trend violations.

While effective at modeling complex temporal patterns, the TSPulse framework has two fundamental limitations. First, the high-dimensional nature of modern time-series data gives rise to the *curse of dimensionality* [10], as the presence of many channels containing irrelevant noise can introduce conflicting signals and degrade model performance [11], as shown in Fig. 2.

The second limitation is its static model selection strategy, called *multi-head triangulation*. This process uses a labeled

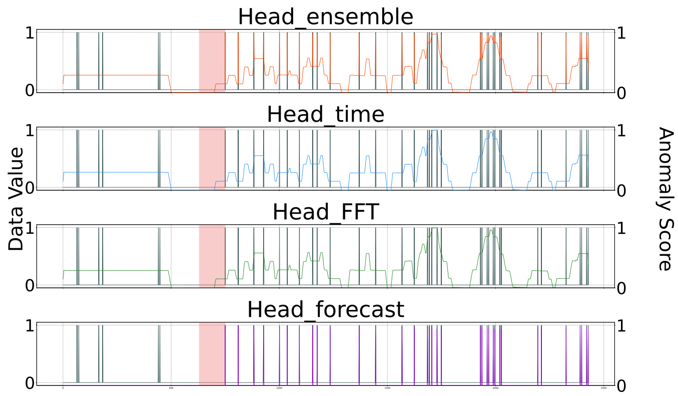

Fig. 2. An uninformative channel from the MSL spacecraft dataset [9]. The raw sensor value (dark slate gray line) remains at zero before and during the anomaly (red shaded region), indicating it was unaffected by the event. As a result, none of the TSPulse heads produce meaningful scores. This demonstrates how irrelevant channels can introduce noise.

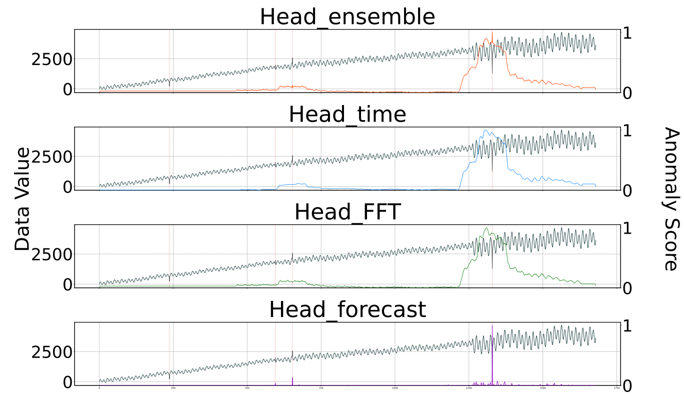

Fig. 3. A visual comparison of anomaly scores from the four TSPulse heads for a sample time series. Each subplot displays the raw data (dark slate gray) and the score from a different head. The ground-truth anomalies are indicated by the thin vertical red lines. This example illustrates how visualization simplifies the task of selecting the best detector. The `Forecast` head clearly provides the most accurate and precise anomaly score; its sharp spikes align precisely with the anomalous events, while the other heads produce broad scores that incorrectly flag many surrounding normal points, leading to false positives.

tuning set to determine the single best-performing head for an entire dataset group, which is then statically applied to all series within that group.

This strategy is flawed in two ways. First, its reliance on a labeled tuning set and pre-computed, dataset-specific rules is a significant practical barrier. While a benchmark can create such a set by reserving data, this approach fails for new, unseen datasets where the dataset identity is unknown, or for datasets too small to provide a representative tuning sample.

Second, even when a tuning set is available, this single choice of a single head for an entire dataset is often the wrong one for individual time series within that dataset. Our analysis reveals that the head chosen by triangulation aligns with the actual best-performing head for only 53.89% of the series in the benchmark, demonstrating the weakness of a static, one-size-fits-all approach.

We propose MAD, a Multimodal Anomaly Detection framework to overcome these limitations. MAD employs a two-stage process designed to tackle both high dimensionality and static model selection.

First, MAD introduces a selective dimensionality reduction stage to address the noise issues of high-dimensional data. This stage focuses the analysis on the most informative data channel instead of processing every channels, using either a method to pre-select the single best channel based on a tuning set or Principal Component Analysis (PCA) [12], [13].

Second, to solve the problem of static model selection, MAD replaces TSPulse's rigid triangulation rule with a dynamic LMM-guided head selection mechanism. Our core insight is inspired by the human practice of visualizing complex data to intuitively spot patterns. As shown in Fig. 3, this visual approach makes the selection task remarkably simple. This observation motivates our use of Large Multimodal Models (LMMs) [14], which are uniquely capable of reasoning over both visual and textual data. The visual modality is also a practical necessity, as visualizations serve to compress lengthy time-series data that would otherwise exceed a language model's context window.

The computational and monetary costs of LMMs are a justifiable trade-off for the paramount accuracy required in high-stakes domains like healthcare [15], [16] and aerospace. Our framework's constrained visual task and deterministic settings ensure reliable, reproducible results, yielding a significant 21.1% relative accuracy improvement over the baseline.

The main contributions of this work are:

- We propose MAD, a novel two-stage framework that synergistically integrates selective dimensionality reduction with a new paradigm for LMM-based dynamic model selection. While the components (PCA, LMMs) are known, their application in this structured process for zero-shot adaptation is novel. We reframe the task not as direct detection, but as a visually grounded classification problem, leveraging plots as an intermediate representation to guide model choice without gradient-based training.
- We achieve state-of-the-art performance on the TSB-AD multivariate benchmark. MAD attains an average VUS-PR score of 0.437, demonstrating a significant 21.1% relative improvement over the zero-shot TSPulse baseline and outperforming other leading methods, all without requiring gradient-based training on the target data.
- We provide a comprehensive empirical analysis, including extensive ablation studies that validate the critical roles of both the selective dimensionality reduction and the dynamic LMM-guided selection stages. Our analysis demonstrates that MAD's adaptive, dataset-agnostic approach is more robust and generalizable than static strategies that rely on impractical prior knowledge of the dataset.

II. The MAD Framework

MAD is a two-stage pipeline designed to enhance both efficiency and detection performance as shown in Fig. 4.

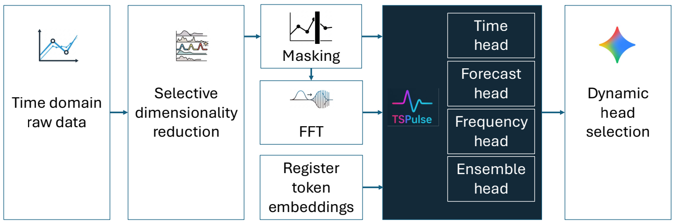

Fig. 4. The MAD Framework Pipeline. (1) A high-dimensional multivariate time series is input. (2) The selective dimensionality reduction stage reduces it to one or more informative components. (3) Anomaly scores are generated for each component using all four TSPulse heads. (4) For each component, a composite plot is created, and an LMM visually selects the most effective head. (5) The score from the LMM-selected head is used as the final anomaly score for that component. Final series scores are averaged if multiple components exist.

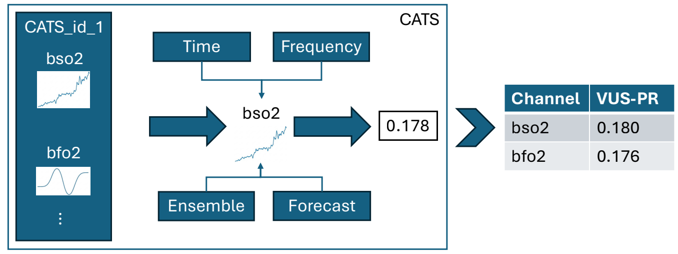

Fig. 5. Schematic of the Best Channel Selection process using the CATS dataset as an example. 1) Each channel (e.g., `bso2`, `bfo2`) from every series in the tuning set is scored by averaging the VUS-PR from all four TSPulse heads. 2) Scores are aggregated per channel name across the dataset group. 3) The channel with the highest final average score (`bso2`) is selected as the most informative for the entire CATS group.

In the first stage, the framework improves signal quality through selective dimensionality reduction, reducing a high-dimensional series to its most informative components using either best channel selection or PCA. This stage is crucial because applying LMM-guided selection to every channel is impractical. After this reduction, the second stage leverages an LMM to perform dynamic, per-channel head selection. The following subsections detail each stage of this process.

A. *Selective Dimensionality Reduction*

It is important to clarify that the underlying TSPulse model, with its specialized heads, is pre-trained. The selective dimensionality reduction stage is not part of the model's training process. Instead, it is an offline analysis performed once on a representative tuning set to define an efficient and robust *inference strategy* for handling new, high-dimensional time series.

The primary approach is to pre-select the single most informative channel, which is applicable when a representative tuning set is available and a consistent channel structure is shared across time series. However, this strategy is not universally applicable. It cannot be used for unseen datasets or those with inconsistent channel structures. For example, our analysis of the `MSL` dataset [9] reveals channel `0` as most informative, but this rule is useless for the `SWaT` dataset [17], which has a different structure. Similarly, within the `MITDB` dataset [18], some series use `MLII` channels while others use `V1`.

To handle these cases, our framework adapts by employing Principal Component Analysis (PCA) as a robust fallback. This approach is motivated by our goal to create a universally applicable framework, one that functions effectively even in scenarios where strategies dependent on specific prior knowledge of the dataset would fail.

1) Best Channel Selection: Our primary strategy is to pre-select the single most informative channel when applicable. An *informative* channel is one where anomalous events produce signals that TSPulse detectors can effectively capture, resulting in a high VUS-PR score. A channel that consistently yields high scores is thus considered informative for our task. This is done through a one-time, offline analysis, detailed conceptually in the steps below and illustrated schematically in Fig. 5, which evaluates all individual channels across all multivariate series in the tuning data.

1. Score Individual Channels: The algorithm iterates through every channel of every time series in the tuning data. For each channel, treated as a univariate series, it first generates four separate anomaly scores using the pre-trained TSPulse heads. From these, it calculates a single composite *informativeness* score by averaging the VUS-PR values across all four heads.

This averaging strategy is a deliberate design choice supported by empirical evidence. We evaluated two approaches for creating the composite score: taking the maximum score from the best-performing head and averaging the scores across all heads. Our experiments on the tuning set revealed that averaging consistently led to the selection of more robustly informative channels, ultimately resulting in better downstream performance.

2. Aggregate Scores by Group: The composite scores for each channel are then aggregated at the dataset group level (e.g., `MSL`, `SMAP` [9]). As illustrated for the `CATS` dataset [19] in Fig. 5, the algorithm calculates the average composite score for each channel name (e.g., `bso2`, `bfo2`) across all series within the tuning set for that group.

3. Select Best Channel: After processing all tuning data, the algorithm calculates the final average score for each channel name within each group. The channel name with the highest average score is designated as the *best channel* for that entire dataset group, creating the final inference map.

This offline analysis produces a simple and efficient mapping from a dataset group to its pre-selected best channel. At inference time, for a new multivariate series, our framework simply looks up its group in this map and processes only the single, pre-selected channel, drastically reducing the computational load while focusing the analysis on the most promising data channel.

2) Principal Component Analysis: In scenarios where best channel pre-selection is not feasible, our framework employs Principal Component Analysis (PCA) as an effective fallback strategy. Our choice is based on the fact that PCA is a well-established technique for both dimensionality reduction and anomaly detection [20]. Its core principle is to project data onto a lower-dimensional subspace that captures the greatest variance. Since anomalies are significant deviations from normal patterns, they often correspond to these high-variance components and are therefore preserved in the principal components. This makes PCA an ideal method for our goal of reducing dimensionality while improving anomaly detection performance. As our own ablation studies confirm (Section III-D1), integrating PCA as a fallback significantly improves detection performance over a no-reduction strategy on the relevant datasets, validating our design choice.

The process begins by standardizing the input data, scaling each of the N channels to have a zero mean and unit variance to ensure that all channels contribute equally to the analysis. The principal components are then derived from the standardized data, and the data is projected onto the subspace spanned by the top k components. The number of components k was treated as a hyperparameter and tuned on the tuning set to determine the optimal output dimensionality, ensuring a robust fallback mechanism. If a time series is reduced to $k > 1$ components, each component is treated as an independent univariate series for the next stage.

B. LMM-Guided Dynamic Head Selection

After reducing the dimensionality of the input, the second stage of MAD is to dynamically select the most appropriate anomaly detection head for the given time series. Building on the insight that this selection is highly intuitive when visualized (Fig. 3), we employ an LMM, which is uniquely capable of processing both visual and textual data to make this choice.

However, a key challenge with LMMs is ensuring their decisions are reliable and not influenced by hallucinations or irrelevant artifacts. To maximize trustworthiness, we designed our framework around three core principles: (1) constraining the task to a four-class classification based on visual evidence, (2) ensuring the LMM provides a clear rationale for human oversight, and (3) enforcing deterministic outputs by setting the model's temperature to 0.

Our visual analysis also revealed a recurring pattern: the `Time` and `FFT` heads often produce visually similar anomaly scores, while the `Forecast` head generates scores with markedly different characteristics. This suggested the selection problem could be framed as a simpler visual classification task, which led us to adopt a *process of elimination* strategy for our few-shot prompt.

We provide the LMM with a set of anonymous examples from the tuning data where the `Time`, `FFT`, or `Ensemble` heads performed well. The LMM is then prompted to choose one of these three heads if the current time series's score plots are visually similar to the provided examples. If they appear *different*, it is instructed to select `Forecast`.

This approach intentionally biases the LMM by defining a visual *in-group* (`Time`/`FFT`/`Ensemble`). This pragmatic design transforms a complex, multi-class visual problem into a more robust *match vs. no-match* decision, leveraging the LMM's pattern recognition capabilities in a constrained and reliable manner.

Crucially, the few-shot examples provided to the LMM are anonymous; the model does not know which dataset they originate from. It makes its selection based purely on the visual characteristics of the time series and anomaly scores at hand, rather than relying on dataset-specific priors. This *dataset-agnostic* approach is far more powerful and applicable to scenarios where an analyst is faced with a novel unseen dataset.

III. Experiments

To evaluate the effectiveness of our proposed framework, we conduct a series of experiments on a standard multivariate time-series anomaly detection benchmark. The experiments aim to validate our design choices and compare MAD's performance against state-of-the-art models.

A. Experimental Setup

We conduct experiments on the comprehensive TSB-AD-M (multivariate) benchmark [3]. To ensure a rigorous evaluation, we adopt the Volume Under the Surface for Precision-Recall (VUS-PR) as our primary metric [6]. This choice aligns with modern best practices in TSAD benchmarking [3], which address known flaws in traditional, threshold-dependent metrics [21]. VUS-PR is robust because it evaluates a detector's performance across all possible anomaly score thresholds and is less sensitive to the precise labeling of range-based anomaly boundaries, making it a more reliable and comprehensive measure of performance.

TABLE I
Dimensionality Reduction Strategy per Dataset. 'BCS' denotes Best Channel Selection, determined from the tuning set. 'PCA' denotes Principal Component Analysis, used as a fallback.

Dataset	Strategy	Details (Selected Channel / k)
CATSv2 [19]	BCS	bso2
CreditCard [22]	PCA	k=3
Daphnet [23]	PCA	k=3
Exathlon [24]	BCS	1-executor-threadpool-activeTasks-value
GECCO [25]	PCA	k=3
GHL [26]	BCS	dL-rand
Genesis [27]	PCA	k=3
LTDB [18]	BCS	ECG1
MITDB [18]	BCS	V1
MSL [9]	BCS	0
OPPORTUNITY [28]	BCS	AccelerometerLAZYCHAIRaccY
PSM [29]	PCA	k=3
SMAP [9]	BCS	0
SMD [30]	BCS	11
SVDB [31]	BCS	ECG1
SWaT [17]	PCA	k=2
TAO [32]	BCS	col-0

Our framework utilizes Gemini 2.5 Pro [33] as the LMM and the TSPulse model. For PCA, the number of components k, upper-bounded by the number of original channels, was determined through hyperparameter tuning for each dataset group. This process resulted in an optimal value of $k = 3$ for most datasets, while for others, such as SWaT ($k = 2$), different values were found to be optimal. A detailed breakdown of the specific dimensionality reduction strategy applied to each dataset is provided in Table I. The visual inputs for the LMM are standardized composite images. Key plotting parameters, such as a large figure width and font size, were empirically tuned to ensure both high temporal resolution in the plots and reliable text recognition by the LMM. All experiments were executed on a single NVIDIA A100-80GB GPU with a fixed random seed and deterministic LMM settings (temperature=0) to ensure reproducibility.

B. Main Results

We compare our framework against three classes of baselines: the foundational pre-trained model (TSPulse), other prominent deep learning models, and a classical statistical method. Our primary baseline is TSPulse, for which we evaluate both the zero-shot (ZS) and fine-tuned (FT) configurations. The ZS version uses the pre-trained model directly and the FT version adapts the model via self-supervised training on a short, anomaly-free segment from each series. Our comparison includes several other prominent deep learning models: Transformer-based architectures such as TranAD [34], TimesNet [35], and AnomalyTransformer [36]; the recently proposed xLSTMAD [37]; and a CNN-based model [2]. Finally, to represent classical statistical methods, we include STL_AD. Following the telemetry analysis approach in recent work [38], this method uses Seasonal-Trend decomposition (STL) [39] on each channel. Anomalies are then identified from the residuals using the Extreme Studentized Deviate (ESD) test [40]. The final score for a series is the mean across all channels.

As shown in Table II, MAD achieves a state-of-the-art average VUS-PR score of 0.437. This represents a 21.1% relative improvement over our primary baseline, the zero-shot TSPulse (0.361), and demonstrates MAD's superiority on 10 of the 17 datasets. Crucially, this performance gain is achieved without any gradient-based training on the target data, validating our core goal of improving zero-shot performance by replacing TSPulse's static selection rule. The framework's adaptive nature is particularly beneficial on complex datasets like `MSL`, `OPPORTUNITY` [28], and `SMAP`, where it substantially outperforms all other methods. While models like CNN excel on specific datasets, MAD's extensible design allows for their future integration as additional candidate heads.

C. Qualitative Analysis: Visual Head Selection

To illustrate MAD's core mechanism, consider the time series from the SMAP dataset in Fig. 6. TSPulse's static triangulation rule selects the `Ensemble` head for the entire SMAP dataset. However, for this specific series, the `FFT` head (VUS-PR 0.918) is over 41% more accurate than the `Ensemble` head (0.649). The reason is visually obvious: the `Forecast` head produces noisy, ineffective scores, which degrades the `Ensemble` score (the max of the other three). MAD avoids this by presenting the anomaly score plots directly to an LMM. Because our framework prompts the LMM to provide a rationale, we can analyze its decision-making process. In this case, it selected `Time` based on visual patterns like *sharp dips* and *cleanly highlighting spikes*, while dismissing the `Forecast` score as being *a mess*. This demonstrates how a dynamic, visual approach, combined with the LMM's interpretable reasoning, allows the model to select the best tool for the specific data and achieve higher performance.

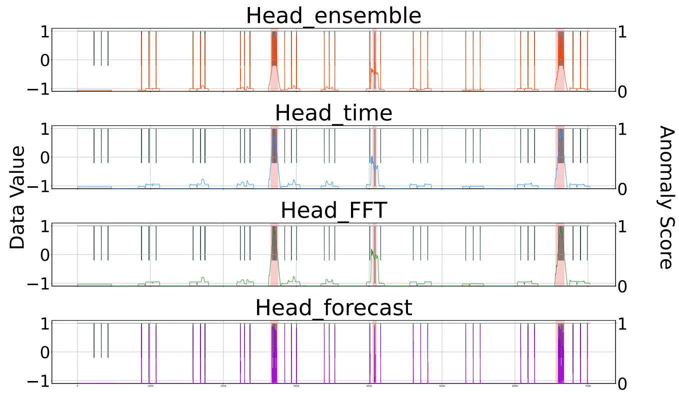

Fig. 6. Qualitative example (SMAP). Dark slate gray: raw signal; colored curves: per-head anomaly scores. The red region (ground truth) is shown only for the reader and is not provided to the LMM. Here, FFT/Time clearly dominate, explaining why dataset-level triangulation (Ensemble) is suboptimal for this channel.

D. Ablation Studies

To validate our framework's core components, we perform targeted ablation studies on the dimensionality reduction and head selection stages.

1) Impact of Dimensionality Reduction: To validate the critical role of our selective dimensionality reduction stage, we conducted an ablation study comparing our full framework (*Full*) against a version where this stage was removed (*Ablated*). In the ablated version, LMM-guided head selection was performed on every single channel of the original high-dimensional time series.

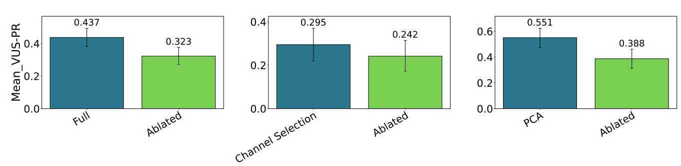

Fig. 7. Effect of dimensionality reduction. Left: overall VUS-PR (Full vs. Ablated). Center/Right: per-subset gains from Best-Channel and PCA ($k = 3$) over "no reduction." Reduction removes uninformative channels and consistently improves detection.

First, as shown in Fig. 7, the dimensionality reduction stage significantly improves detection performance, boosting the average VUS-PR score from 0.323 for the ablated version to 0.437 for the full framework. A granular analysis confirms this: on the subset of data where Best Channel Selection is applicable, its average VUS-PR of 0.295 (95% CI: [0.220, 0.370]) outperforms the ablated baseline's 0.242 (95% CI: [0.173, 0.314]). The improvement is even more pronounced on the subset requiring PCA, where its score of 0.551 (95% CI:

TABLE II

VUS-PR ON TSB-AD-M (HIGHER IS BETTER). BOLD MARKS THE BEST PER DATASET AND THE BEST OVERALL AVERAGE. TSPULSE (FT) AND XLSTMAD RESULTS ARE FROM THEIR ORIGINAL PAPERS; OTHER BASELINES ARE FROM TSB-AD OR OUR REPRODUCTIONS. MAD IS OUR METHOD. WE REPORT THREE DECIMALS THROUGHOUT FOR RESULTS FROM TSB-AD OR INTERNAL RESULTS.

Dataset	Ours	TSPulse (FT)	TSPulse (ZS)	xLSTMAD	TranAD	CNN	TimesNet	AnomalyTransformer	STL_AD
CATSv2	0.053	0.07	0.055	**0.10**	0.042	0.082	0.068	0.035	0.063
CreditCard	0.007	0.00	0.004	**0.06**	0.022	0.022	0.020	0.020	0.049
Daphnet	0.357	0.35	0.354	**0.50**	0.311	0.208	0.275	0.072	0.412
Exathlon	0.862	**0.91**	0.895	**0.91**	0.096	0.681	0.421	0.102	0.756
GECCO	0.150	0.18	0.177	0.04	0.020	0.033	0.035	0.022	**0.253**
GHL	0.022	0.01	0.007	0.01	**0.056**	0.024	0.008	0.028	0.011
Genesis	0.007	0.02	0.010	0.01	0.036	**0.101**	0.019	0.014	0.003
LTDB	0.443	**0.57**	0.359	0.40	0.258	0.328	0.272	0.214	0.241
MITDB	0.082	**0.14**	0.073	0.09	0.073	**0.140**	0.068	0.051	0.081
MSL	**0.618**	0.21	0.202	0.38	0.242	0.352	0.172	0.116	0.202
OPPORTUNITY	**0.464**	0.07	0.062	0.16	0.165	0.158	0.055	0.072	0.081
PSM	0.140	0.14	0.140	0.18	**0.232**	0.219	0.140	0.209	0.131
SMAP	**0.605**	0.32	0.304	0.35	0.087	0.191	0.094	0.063	0.141
SMD	0.192	**0.36**	0.354	0.35	0.301	0.353	0.142	0.067	0.154
SVDB	0.399	**0.47**	0.385	0.26	0.119	0.187	0.111	0.082	0.116
SWaT	0.105	0.14	0.128	0.34	0.154	**0.405**	0.136	0.181	0.148
TAO	0.978	0.93	0.933	0.80	0.808	**0.998**	0.786	0.770	0.903
Average	**0.437**	0.39	0.361	0.37	0.199	0.313	0.166	0.119	0.220

[0.474, 0.623]) is a substantial gain over the ablated score of 0.388 (95% CI: [0.314, 0.461]). These results confirm that our adaptive reduction strategy consistently improves performance.

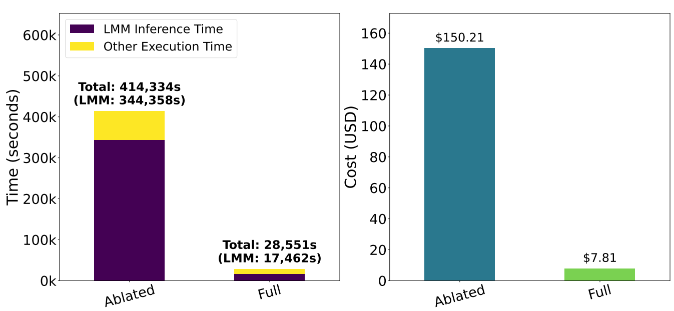

Fig. 8. Efficiency with/without reduction. Left: >93% drop in total time; right: >94% drop in API cost when reducing before LMM selection.

Second, this stage is crucial for efficiency. As illustrated in Fig. 8, dimensionality reduction reduced the total execution time from over 414,000s to approximately 28,500s (a reduction of over 93%) and the estimated API cost from over $150 to just $7.81 (a reduction of over 94%) compared to the ablated approach. These results confirm that processing every channel is computationally infeasible, making the DR stage a critical component for achieving state-of-the-art performance in a practical manner. The time required for the one-time offline analysis to determine the best channels or PCA parameters is negligible, as it operates on the small tuning set (20 series) and consists mainly of running the baseline TSPulse model once. This setup cost is amortized over all subsequent inferences.

While providing significant efficiency gains, our dimensionality reduction strategy also introduces a trade-off between computational cost and accuracy, particularly in the selection of the number of PCA components, k. To analyze this, we evaluated the impact of varying k on the tuning data, with results summarized in Table III. Using only the first principal component ($k = 1$) resulted in the lowest performance, indicating that crucial anomaly information is not always captured by the component with the highest variance. Increasing to $k = 3$ yielded the optimal VUS-PR score of 0.331. While this represents a 200% increase in computational cost relative to $k = 1$, the significant accuracy gain justifies the trade-off. Further increasing to $k = 4$ showed diminishing returns, with a higher cost for lower accuracy. Our chosen value of $k = 3$, determined via this tuning process, therefore represents an empirically validated balance between cost and accuracy.

TABLE III

COST-ACCURACY TRADE-OFF FOR PCA COMPONENTS (k) ON TUNING SET

k	VUS-PR (PCA Subset)	Relative Cost
1	0.265	**1.0x**
2	0.304	2.0x
3	**0.331**	3.0x
4	0.289	4.0x

2) Impact of Head Selection Strategy: To isolate and validate the contribution of our dynamic, LMM-guided head selection, we conducted a comprehensive ablation study. Our framework (MAD) operates on a per-channel basis after dimensionality reduction. We compare our main few-shot LMM approach against several key baselines: 1) A zero-shot version of our LMM-guided selection. 2) The static *multi-head triangulation* strategy, which uses a tuning set to pick the best head for the dataset. 3) A theoretical *Series-Level Optimal* baseline, which represents the upper-bound performance achievable by selecting the single best head for each series post-hoc. 4) Static fixed-head strategies to illustrate the risk of a poor a priori choice.

We define a theoretical *Series-Level Optimal* baseline (selecting the best head per series post-hoc), as a true *Channel-Level Optimal* selection is computationally infeasible (4^n combinations for n channels). An attempt to adapt the TSLens [8] attention mechanism to learn a selection function was unsuccessful, as it failed to generalize beyond the most frequent choice in the tuning data.

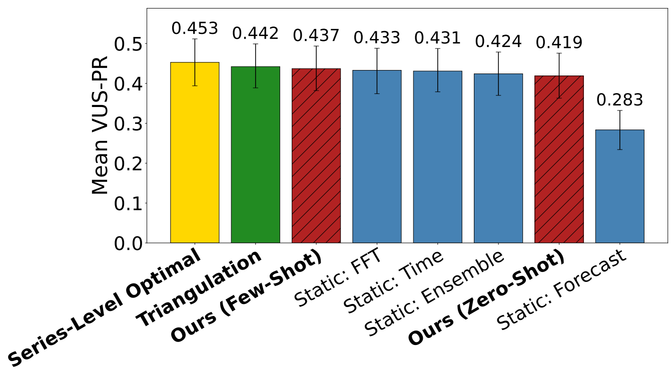

Fig. 9. Head-selection strategies on TSB-AD-M. Dataset-specific triangulation attains 0.442 but requires a labeled tuning set and dataset identity. MAD's few-shot, dataset-agnostic selection reaches 0.437, providing a practical and statistically competitive alternative for unseen datasets without prior metadata. Fixed-head baselines illustrate the risk of static choices.

The results, detailed in Fig. 9, reveal a crucial trade-off between benchmark-specific performance and real-world generalizability. While the static *multi-head triangulation* strategy achieves the highest score among practical methods (0.442), its reliance on a labeled tuning set and dataset identity makes it impractical for many real-world scenarios. In contrast, our primary method, MAD, achieves a statistically competitive VUS-PR of 0.437 (95% CI: [0.382, 0.494]) by operating under a more challenging and realistic *dataset-agnostic* assumption, as detailed in Section II-B.

To validate our chosen *process of elimination* prompting strategy (Section II-B), we evaluated several alternatives on the tuning set. A strategy that provided examples only where `Forecast` was optimal and asked the LMM to select it for visually similar cases performed poorly (VUS-PR 0.329). A more comprehensive approach with examples for every possible best head (`Time`, `FFT`, `Forecast`, and `Ensemble`) was more effective but still suboptimal (0.418). Our final strategy (0.437) proved the most robust, confirming that framing the task as a *match vs. no-match* decision is highly effective.

Furthermore, the fixed-head baselines highlight the significant risk of static choices. A good guess, like `FFT` (0.433), performs well, while a poor one, like `Forecast` (0.283), leads to a catastrophic performance collapse. Our LMM-guided method successfully mitigates this risk by adapting to the data. According to the Wilcoxon signed-rank test, MAD is statistically indistinguishable from the best static choice on this benchmark (`FFT`, $p = 0.36$) while significantly outperforming the other fixed-head strategies: `Time` ($p = 0.016$), `Ensemble` ($p = 0.014$), and `Forecast` ($p < 0.001$). This confirms it provides a robust selection without dataset-specific tuning. The small performance gap between our method and the triangulation baseline is a worthwhile trade-off for a system that is fundamentally more adaptive and generalizable.

IV. Conclusion

We introduced MAD, a multimodal framework that couples selective dimensionality reduction with LMM-guided, per-channel head selection. This design addresses two practical bottlenecks of pre-trained TSAD: high-dimensional noise and static, dataset-level head choice. On TSB-AD-M, MAD achieves a mean VUS-PR of 0.437, outperforming TSPulse (ZS 0.361) and xLSTMAD (0.37) without training. It remains competitive with dataset-specific triangulation (0.442) while offering a more generalizable, dataset-agnostic approach suitable for real-world deployments where prior dataset knowledge is unavailable. Ablations confirm that the reduction stage delivers large efficiency gains (>93% time, >94% cost reductions), enabling practical LMM use at scale.

Limitations and Outlook. Our framework has limitations: the best-channel selection assumes stable channel naming, PCA is not universally optimal, and reliance on a closed-source LMM introduces reproducibility and bias concerns. To mitigate the latter, we are releasing all prompts, scripts, and seeds. Future work must validate MAD's generalization on real-world streaming data and address domain drift, where performance may degrade over time. A critical research direction is the development of lightweight, self-adaptive selectors (such as smaller, specialized vision models or non-LMM heuristics) to reduce dependency on large external APIs. We also plan to integrate diverse detectors like CNNs as additional candidate heads and conduct robustness studies on plot-style perturbations.

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